Databases Project

Choose your adventure

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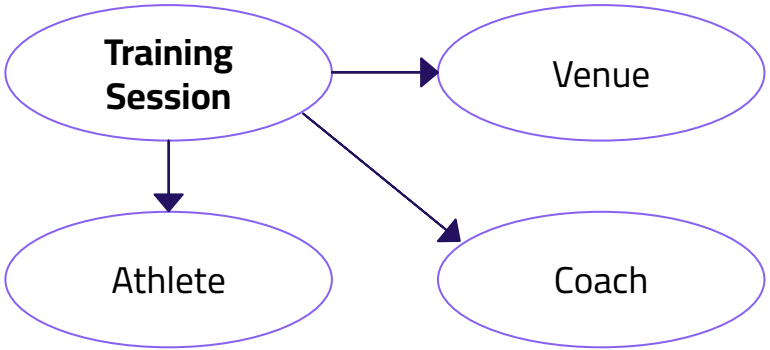
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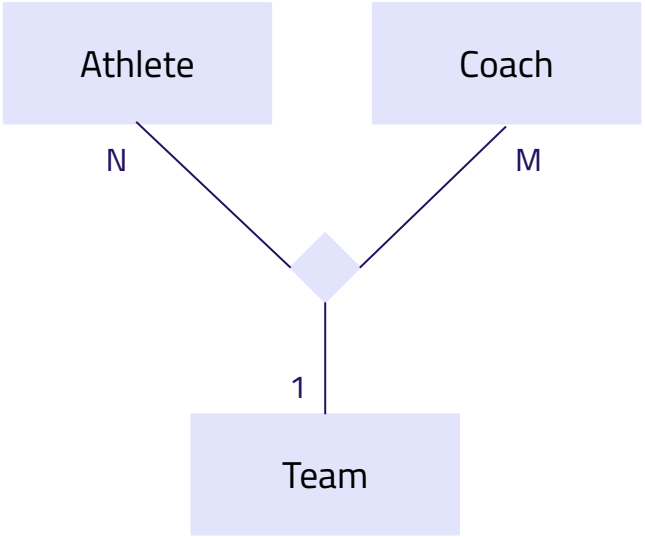
Dataset Description

- The data came fragmented across multiple CSV files.
- Some files had duplicated information or indirect relationships.
- The additional CSV implemented, expanded the approach.



athletes.csv	teams.csv
coaches.csv	venues.csv
technical_officials.csv	schedules.csv
medals.csv	nocs.csv
medallists.csv	events.csv
training_sessions.csv	

Design Decisions



1. **Many to Many Relationships**
   Less duplicated data, easier updates, better scalability.

2. **Normalization**
   Removed transitive dependencies: NOC data separated from Athletes, Venue data separated from Schedules to avoid duplication.

3. **Data Type Choices**
   Dates: DATE
   Country codes: CHAR(3)
   Athlete names: VARCHAR2(100)
   Medal types: VARCHAR2(10)

Design Decisions

Trigger 1	Trigger 2	Trigger 3
Warning Trigger	Athlete Status	Medal History Trigger
BEFORE INSERT OR UPDATE	AFTER INSERT	AFTER INSERT
Insertion is Non-Blocking	If 3 training sessions with an intensity higher than 6	Converts medal types into strings using CASE
Warning Trigger	Medal History Trigger	Validation Trigger

TRAINING_ID	MESSAGE
3000000	Training date is NULL

ATHLETE_CODE	IS_CURRENT
1878715	True

SECOND_NAME	MEDAL_TYPE	MEDAL_LABEL
CHIA	Gold Medal	GOLD

QUERY 1 - Countries with gold, silver and bronze medals

Which countries have managed to win gold, silver and bronze medals in individual competitions?

```
SELECT c.country
FROM Country c
JOIN Person p ON c.country_code = p.nac_code
JOIN Athlete a ON p.person_code = a.athlete_code
JOIN Athlete_wins w ON a.athlete_code = w.athlete_code
JOIN Medal m ON w.medal_id = m.medal_id
WHERE w.is_medallist = 'True'
GROUP BY c.country
HAVING SUM(CASE WHEN m.medal_type = 'Gold Medal' THEN 1 ELSE 0 END) > 0 AND
SUM(CASE WHEN m.medal_type = 'Silver Medal' THEN 1 ELSE 0 END) > 0 AND
SUM(CASE WHEN m.medal_type = 'Bronze Medal' THEN 1 ELSE 0 END) > 0;
```

	COUNTRY
1	IR Iran
2	Georgia
3	Poland
4	Thailand
5	Israel
6	United States
7	Argentina
8	Croatia
9	Norway
10	Brazil
11	Azerbaijan

Only 11 countries achieved gold, silver and bronze medals in individual events, with the United States clearly leading the ranking.

QUERY 2 - Venues hosting events from multiple sports

Which venues have hosted events belonging to more than one different sport?

```
SELECT v.venue_name, s.sport, COUNT(DISTINCT sch.event_ID) AS events
FROM Venue v
JOIN Schedule sch  ON v.venue_code = sch.venue
JOIN Showcases swc ON sch.event_ID = swc.event_ID
JOIN Sport s ON swc.sport_code = s.sport_code
WHERE v.venue_name <= ALL (
            SELECT v2.venue_name
            FROM Venue v2
            JOIN Schedule sch2 ON v2.venue_code = sch2.venue
            JOIN Showcases swc2 ON sch2.event_ID = swc2.event_ID
            GROUP BY v2.venue_name
            HAVING COUNT(DISTINCT swc2.sport_code) > 1
)
GROUP BY v.venue_name, s.sport;
```

	VENUE_NAME	TOTAL_SPORTS
1	Yves-du-Manoir Stadium	12
2	South Paris Arena 1	12
3	Pierre Mauroy Stadium	12
4	South Paris Arena 6	13
5	Aquatics Centre	14
6	Stade de France	13
7	Bordeaux Stadium	12
8	Lyon Stadium	12
9	South Paris Arena 4	5
10	Bercy Arena	15

Out of all Olympic venues, 32 hosted events from multiple sports, with Bercy Arena and the Aquatics Centre standing out as the most versatile.

QUERY 3 - Countries with individual and team medalists

Which countries have both individual medalists and team medalists?

```
SELECT p.first_name || ' ' || p.second_name AS athlete_name
FROM Person p
JOIN Athlete a ON p.person_code = a.athlete_code
JOIN Athlete_wins aw ON a.athlete_code = aw.athlete_code
WHERE aw.is_medallist = 'True'

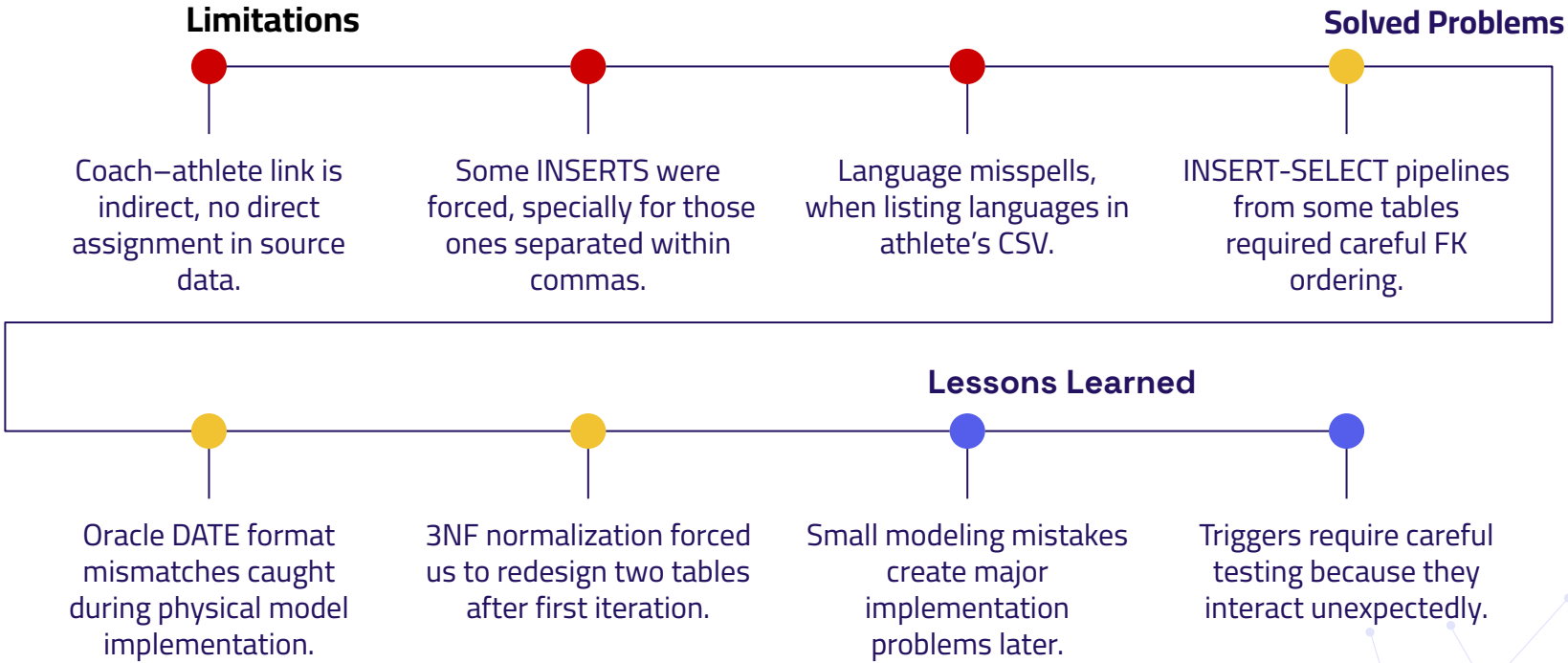
INTERSECT

SELECT p.first_name || ' ' || p.second_name
FROM Person p
JOIN Athlete a ON p.person_code = a.athlete_code
JOIN Members m ON a.athlete_code = m.athlete_code
JOIN Team t ON m.team_code = t.team_code
JOIN Team_wins tw ON t.team_code = tw.team_code
WHERE tw.is_medallist = 'True';
```

	COUNTRY
1	Argentina
2	India
3	Italy
4	France
5	Great Britain
6	Germany
7	Croatia
8	Morocco
9	Denmark
10	Poland
11	Norway
12	South Africa
13	Spain

Only 13 countries won medals in individual and team competitions, with Italy exemplifying this balance.

Conclusions



Bibliography

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Oracle Docs	docs.oracle.com, SQL Language Reference
Data Generation	mockaroo.com, Custom CSV Generation
ER Modeling	miro.com, Collaborative Diagrams
Stack Overflow	https://stackoverflow.com/questions, For Issues & Questions

Thanks!

